

All-Digital PLL for Bluetooth Low Energy Using 32.768-kHz Reference Clock and ≤ 0.45 -V Supply

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Abstract—In this paper, we introduce an all-digital phase-locked loop (ADPLL) for Bluetooth low energy (BLE) that eliminates the need for a crystal oscillator (XO) other than a 32.768-kHz real-time clock (RTC) already present in wireless systems. Specifically, we propose to replace the conventional channel settling with a band settling that would be carried out only once per global device power up. The ADPLL locks to the center of the Bluetooth band (2440 MHz) upon system power-up and jointly performs an instantaneous channel hopping and Gaussian frequency shift keying (GFSK) modulation in a two-point manner to overcome the narrow PLL bandwidth (BW) due to the 32.768-kHz reference. Extensive calibrations linearize the effective cubic digitally controlled oscillator (DCO) transfer function to achieve a precise frequency range of hopping and modulation. Realized in 16-nm FinFET, it consumes <1 mW at ≤ 0.45 V, while achieving best-in-class performance and <100 -ns hopping time.

Index Terms—All-digital phase-locked loop (ADPLL), Bluetooth low energy (BLE), channel hopping, digitally controlled oscillator (DCO), FinFET, real-time clock (RTC), voltage doubler.

I. INTRODUCTION

THE current paradigm of frequency synthesis for short-range wireless transceivers, such as Bluetooth low energy (BLE), is to use a crystal oscillator (XO) in the tens of megahertz range as a frequency reference (FREF) to phase lock an RF oscillator [1]–[10]. The high reference frequency (f_R) ensures a sufficiently wide bandwidth (BW) of tens to hundreds of kilohertz of the surrounding phase-locked loop (PLL) to: 1) quickly acquire a new channel, preferably with loop presetting techniques [19], [28], [29] and 2) suppress lower frequency phase noise (PN) of the RF oscillator. When exploiting the latest advancements in the field of RF all-digital PLLs (ADPLL), the latter requirement can be alleviated by substantially lowering the flicker PN of a digitally controlled oscillator (DCO), thus allowing to freeze its tuning word updates during receive (RX) packets and further directly FM-modulating the DCO during transmit (TX) packets [5]. However, the ADPLL is still needed just

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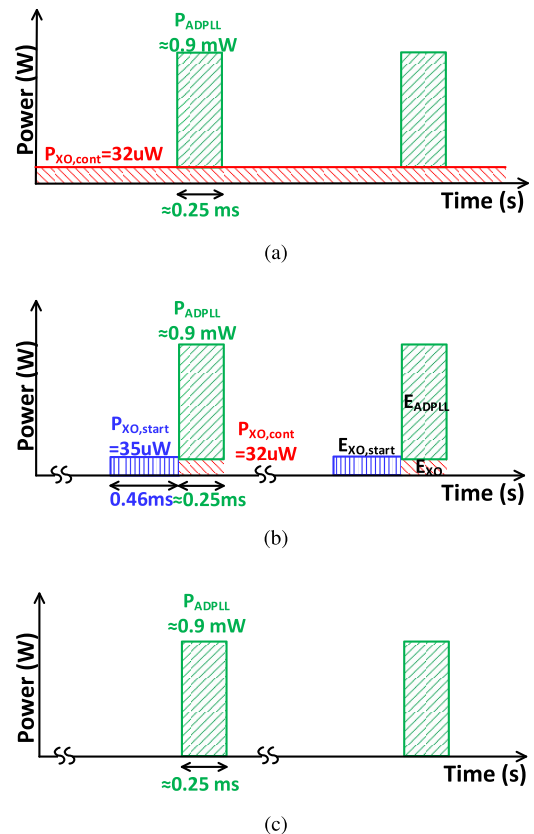


Fig. 1. (a) Continuous XO operation. (b) XO restart before each packet [XO's latest advancements from [26], [27] in (a)–(b)]. (c) Proposed new paradigm (i.e., no XO).

to quickly settle the DCO to each new channel. Afterward, the ADPLL can be stopped. As long as the registers controlling the oscillator tuning word (OTW) can retain their state, the rest of the ADPLL circuitry can be entirely shut down.

The old paradigm serves its purpose well as long as the energy consumption of XO is just a small fraction of the complete RF synthesizer. A typical XO requires around $P_{XO} = 100 \mu\text{W}$ just to continuously sustain itself, although that number went down just recently to $35 \mu\text{W}$ due to novel energy pulse injection techniques [26], [27] [see reflected in Fig. 1(a)]. The lower power limit is dictated by the crystal's quality (Q)-factor and the required signal-to-noise ratio at a given oscillating amplitude, which are all difficult

to tradeoff. With the recent advancements in ADPLLs for IoT RF transceivers, the power consumption of RF synthesizers, P_{PLL} , now reaches a sub-milliwatt level [4], [6]–[10]. The power ratio, $P_{\text{XO}}/P_{\text{PLL}}$ of ~ 0.04 is still reasonably low if the RF transceiver would operate continuously. However, with the packetized operation of IoT radios, this is not the case. In fact, the IoT radios trend is to rather operate at very low duty cycles, D , as shown in Fig. 1(a). Now, with only $D = 0.04$, the energy consumption of XO, E_{XO} , is at the same level as that of the RF synthesizer, E_{PLL} .

A typical solution to the above-mentioned problem of E_{XO} is to periodically shut down the XO. However, until a couple of years ago, each XO restart would be very energy intensive (~ 1 mW over ~ 1 ms) so only XO shutdowns > 10 ms make sense [11]. In such systems, the energy drained by the XO can now be actually higher than in an RF PLL (~ 1 mW over ~ 0.5 ms). Recent works have dramatically reduced the XO startup time and energy: via dithered frequency injection: $64 \mu\text{s}$ in [25]; by dynamically adjusting load capacitance of its Pierce oscillator: $200 \mu\text{s}$, 36 nJ in [12]; by injecting well controlled and timed energy pulses: $2.2 \mu\text{s}$, 13.3 nJ in [27], and $400 \mu\text{s}$, 14.3 nJ in [26]. The best XO power-up scenario is depicted in Fig. 1(b).

In the spirit of continual quest to reduce energy consumption of IoT transceivers, as well as the overall system cost and occupied volume, we propose to eliminate the need for the high-frequency XO and use a reference of a much lower frequency, such as a 32.768-kHz real-time clock (RTC) that never shuts down as it consumes negligible power, as shown in Fig. 1(c). This drastic clock slow-down allows now for the digital logic circuitry (with f_T of hundreds of gigahertz) to operate at sub-threshold further reducing power consumption. Any needed clocks at megahertz-level frequencies can be derived by down dividing the DCO edges [13]. We verify the technique in an ultralow voltage (ULV) ADPLL realized in 16-nm FinFET CMOS.

The rest of this paper is organized as follows. Sections II and III introduce a new paradigm of RF frequency synthesis for channel hopping of BLE radios. Section IV gives an overview on how to estimate the DCO gain (K_{DCO}) with an LMS-based calibration. Section V describes the DCO architecture and design. Section VI derives the nonlinear K_{DCO} as a cubic function of the resonant frequency and explains how to compensate the nonlinearity. Section VII describes a top-level view of the proposed ADPLL. Section VIII points out relevant challenges of an ULV design. Finally, Section IX demonstrates experimental results.

II. NEW PARADIGM FOR IoT RF PLLs

In this paper, we experimentally demonstrate the elimination of traditional XO that underpins the current paradigm of regular (i.e., tens to hundreds of kilohertz) BW PLL operation for BLE by resorting to a 2^{15} Hz = 32.768-kHz RTC. The RTC has historically made its way into electronic wrist watches, by providing 1-s ticks via a simple 2^{15} division. The RTC is now a ubiquitous component in IoT hosts and is responsible

for TX/RX scheduling. A < 20 -ppm frequency accuracy¹ is now widely available² (see [32]) as it corresponds to < 2 -s timing accuracy per day, which seems rather necessary for time keeping.

The new paradigm obviously invokes new challenges. The first challenge is the resulting very low PLL BW (f_{BW}) of ~ 1 kHz, which is dictated by the well-known PLL stability rule: $f_{\text{BW}}/f_R < 1/10$. The very low BW implies a very sluggish loop response to large perturbations, such as acquiring a new channel during the channel hopping operation. On the other hand, the current phase noise performance of low-power oscillators has reached the level of making the PLL loop unnecessary for just tracking within BLE packets [5]. However, the settling time response to hopping channels still takes tens to hundreds of FREF cycles such that it would be comparable itself to the packet length of $\sim 250 \mu\text{s}$, thus making it impractical. The only workable solution appears in avoiding any uncertainty with the PLL loop components, thus knowing all of them “precisely.” This allows to instantaneously generate new frequency points without the need for subsequent settling, in a manner similar to an open-loop frequency generation. This is in stark contrast with the current paradigm of tolerating significant inaccuracies in the individual PLL component characteristics and allowing the loop settling to work out any such inaccuracies over time. Clearly, the wider the BW is, the shorter is the time.

The process, voltage, and temperature (PVT) variations will naturally affect the component characteristics thus imply imperfect knowledge thereof. The negative feedback loop appears, therefore, necessary in one form or another to compensate for these variations. However, this loop does not need to run at tens to hundreds of kilohertz BW since the PVT variations are slower by several orders of magnitude.

The above-mentioned observation contrasts the current paradigm with the proposed solution in the following way: Instead of ignoring the knowledge of past PLL behavior, effectively throwing that information away and starting anew each time but with a high loop BW, the loop now can retain the previous knowledge such that responses to large deterministic perturbations, e.g., new hopping channels, become error-free thus requiring no loop corrections and no settling.

Implied in the proposed paradigm is digital or digitally intensive nature of the PLL components (e.g., ADPLL) and calibration loop as necessary to ensure the precision. For example, the transfer function of a digital loop filter is accurate (unless bits are inadvertently dropped). The transfer function of a DCO is much easier to estimate and calibrate than that of a voltage-controlled oscillator (VCO) due to its pure digital input which can generate exact stimulus with the rest of the loop digitally detecting its response [1], [7], [15], [23], [24].

Having adopted the ADPLL architecture for its potential to perform the ultra-fast [28], [29] or even instantaneous and accurate frequency hopping which require no settling, the next

¹BLE requires 75 kHz of initial frequency accuracy at the beginning of each packet, which translates to 30 ppm, from which 20 ppm is allocated to the crystal.

²While currently, the prices for 20-ppm RTC are higher than for 100 ppm ones, they are expected to drop with the increased volume [33].

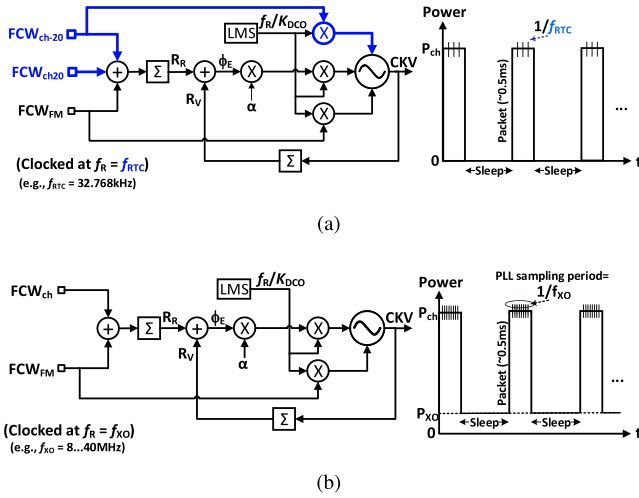


Fig. 2. Simplified block diagram of ADPLL with TPM. (a) Current paradigm and (b) new paradigm for channel hopping in IoT RF PLLs.

issue is the form of that realization. The conventional ADPLL with a two-point modulation (TPM), as shown in Fig. 2(a), is composed of a reference accumulator (producing R_R), phase detector (producing ϕ_E), loop filter (only the proportional coefficient α shown), variable accumulator (producing R_V), DCO gain normalizing multiplier f_R/K_{DCO} , and DCO. Conventionally, each hopping channel, controlled by FCW_{ch} (with frequency command word (FCW) defined as the frequency multiplication by ADPLL [19]), needs to settle in closed loop at high BW. The DCO has two ports feeding separate OTWs for: 1) channel settling and tracking, OTW_{ch} and 2) modulation, OTW_{FM} .

We propose in Fig. 2(b) to replace the conventional channel settling with a *band* settling that would be carried out only once per global device power up. When not in a sleep mode, the ADPLL will always stay locked to the center channel (CH_{20} at 2440 MHz $\Rightarrow FCW_{ch20} = 2440 \text{ MHz}/2^{15} \text{ Hz} = 74462.891$) and perform the combined channel hopping (FCW_{ch-20}) and FM modulation (FCW_{FM}) by instantaneously *offsetting* the DCO resonance from the center channel (CH_{20}) via a TPM. In the FCW_{ch-20} notation, $ch-20$ is the number of $\Delta f_{ch} = 2\text{-MHz}$ channels away from the middle channel 20.³ The DCO has now three ports for: 1) OTW_{ch20} : settling and tracking the center channel 20; 2) OTW_{FM} : modulation; and 3) OTW_{ch-20} : jumping to a new channel from CH_{20} . The formulas in the following capture the relationships for 2) and 3):

$$OTW_{FM} = FCW_{FM} \cdot f_R / K_{DCO} \quad (1)$$

$$\begin{aligned} OTW_{ch-20} &= (ch - 20) \cdot \Delta f_{ch} / K_{DCO} \\ &= [(ch - 20) \cdot \Delta f_{ch} / f_R] \cdot f_R / K_{DCO} \\ &= FCW_{ch-20} \cdot f_R / K_{DCO}. \end{aligned} \quad (2)$$

With the accurate K_{DCO} estimation, the ADPLL loop plays no role there, hence f_{BW} should theoretically not matter. Of course, OTW_{ch20} undergoes the normal ADPLL settling

³For simplicity, $k = 1, 2, \dots, 40$ channel is defined here as $2400 \text{ MHz} + k \cdot 2 \text{ MHz}$

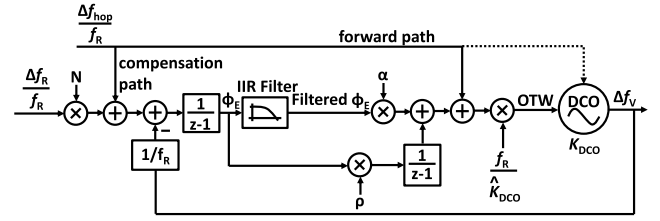


Fig. 3. ADPLL with two-point FM modulation.

and tracking. However, since it is done only once per power up to acquire the fixed frequency, the low $f_{BW} = \sim 1 \text{ kHz}$ seems sufficient.

A. Background of Two-Point Modulation

A TPM of a PLL is arguably the most popular technique for the data FM modulation in short-range wireless radios. It is readily implemented in ADPLLs exclusively using digital logic [14], but analog PLLs require some additional analog circuitry. As the z-domain model of Fig. 3 shows, “two-point” implies: 1) the modulating data are fed directly into the DCO in a feed-forward manner and 2) the compensating data are fed into the phase detector. TPM requires precise knowledge of the DCO gain, K_{DCO} , for the transfer function to be “all-pass.” This way, the loop dynamics are removed from the modulating path, while the noise and error sources undergo the normal closed-loop attenuation. The TPM modulator can tolerate some K_{DCO} error, which is inversely proportional to the modulation BW. For Bluetooth, it is a few percent [1].

III. ACQUISITION OF BLE CHANNEL HOPPING AND MODULATION

Fig. 4(a) is a conceptual plot of oscillating frequency versus time involving both data modulation (exaggerated!) and channel hopping. BLE contains 40 channels, between 2402 and 2480 MHz, separated by 2 MHz [16]. The modulation is Gaussian frequency shift keying (GFSK) with a modulation index $h = 0.5$, giving $\pm 250\text{-kHz}$ frequency deviations for a sequence of symbols “0” and “1,” i.e., with no inter-symbol interference (ISI).

If K_{DCO} -normalized value is precise, the instantaneous hopping and $\pm 250\text{-kHz}$ frequency modulation are achieved. If the K_{DCO} -normalized value is over(under) estimated, longer hopping settling will result and likely to extend over the entire packet, as Fig. 4(b) shows. In this case, the modulation index will be too large, as Fig. 4(c) indicates.

It should be stressed that K_{DCO} accuracy, which also needs to track PVT variations, is now much more critical for the channel hopping than for the GFSK modulation. For a given error in K_{DCO} estimation, the frequency error is proportional to the frequency jump. Hence, a 0.1% K_{DCO} error results in a 250-Hz error for the 250-kHz modulation, whereas it can be as large as 40 kHz for the extreme channel hopping of 40 MHz.

The chip-to-chip process variation is fully compensated for via the power-up PLL settling and LMS adaptation. Thanks to the low-frequency pushing of the adopted

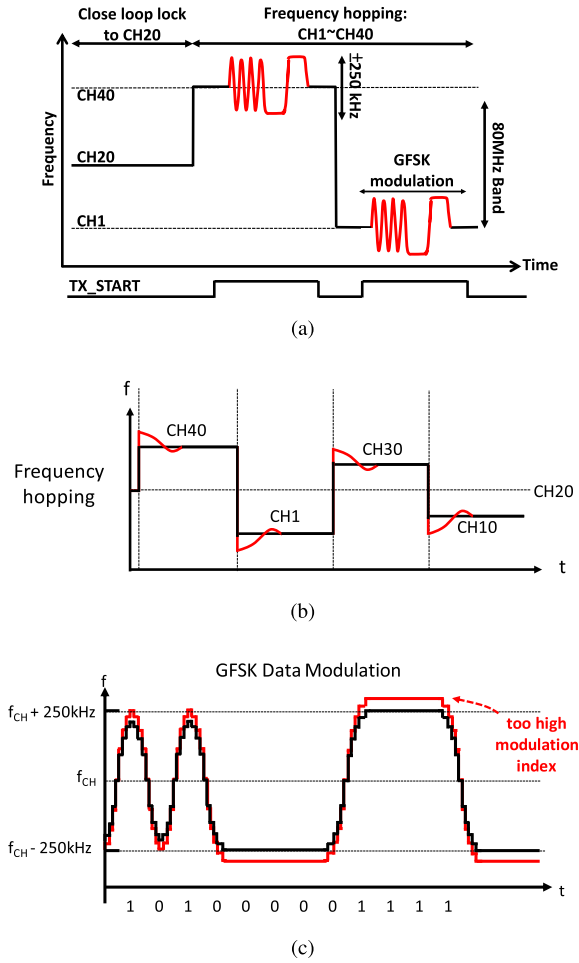


Fig. 4. (a) Combined BLE channel hopping and modulation. (b) Frequency hopping with under estimated K_{DCO} , i.e., f_R/K_{DCO} multiplier is too high. (c) Larger modulation index with under estimated K_{DCO} .

DCO [21], the voltage supply effects would be barely noticeable with a low dropout (LDO) regulator merely providing a 30–40-dB rejection. As for the temperature induced changes, the DCO is stable enough to tolerate 4 s of the maximum allowed separation between transmitted packets (so-called connection interval is specified to be between 7.5 ms and 4 s). The maximum rate of temperature change inside a cellular phone, occurring due to sudden heating up of the host's RF power amplifier, should be less than 1°C/s . That rate is expected to be much lower in an IoT environment. To be able to perform a proper channel hop upon a short sleep, the frequency shall drift be less than 75 kHz in that period and the DCO gain shall develop less error than $75\text{ kHz}/40\text{ MHz} = 0.18\%$. In this design, the DCO frequency and gain drifts were measured, respectively, at 1.66% and 1.73% between 25 and 100°C , translating into lower than 0.1% drift in the 4°C range.

Naturally, when waking up from a prolonged shutdown, the DCO can experience the full temperature change. In that case, either 1) an additional dummy packet be allocated to the DCO settling and calibration or 2) lookup table compensation performed based on temperature sensor readings.

IV. DCO GAIN ESTIMATION

In [14], the DCO gain normalizing multiplier f_R/K_{DCO} was calibrated via an LMS algorithm that correlated zero-crossing directions of the modulating data with the resultant filtered phase error while driving that correlation to zero. When that condition is reached, the TPM BW would be all-pass. The calibration was carried out entirely within each packet with necessary resets in-between (each channel would have a different value of K_{DCO}). Hence, the prior packets would not benefit the estimation in the current packet. In this paper, we adopt that LMS-based calibration method to accurately estimate the DCO gain but with the following modifications: 1) the ADPLL is settled at only one frequency (i.e., at the center of Bluetooth band, 2440 MHz), hence avoiding the issues with K_{DCO} being channel dependent; 2) the LMS input data are not the modulating data but channel hopping perturbation at the beginning of each packet; 3) the LMS adaptation starts upon the system restart and continues without resetting until the system shutdown. The much higher extend of the hopping perturbations compensates for their slower rate.

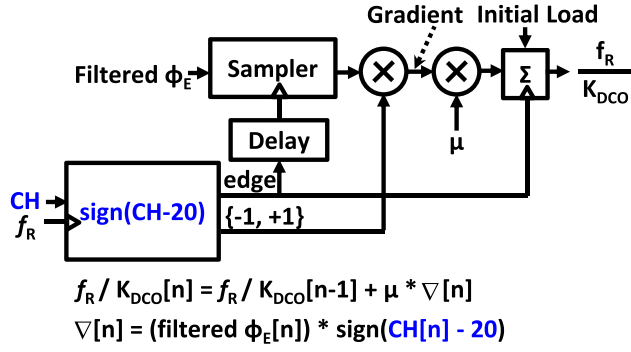
The LMS adaptation loop, whose critical part is shown in Fig. 5, observes the filtered phase error ϕ_E upon a change of the channel selection CH to determine whether the TPM BW is all-pass. If not, based on the well-known LMS equations, the gradient ∇ , i.e., the correlation between the channel direction $\text{sign}(\text{CH}-20)$ and filtered ϕ_E , is scaled down by the step-size μ and added to the current normalization value of the DCO gain multiplier, f_R/K_{DCO} . The calibration procedure continues to adapt while its phase error goes toward zero. When settled, the estimated f_R/K_{DCO} is very close to the actual value of f_R/K_{DCO} .

It should be noted that for other IoT standards, where channel hopping could be infrequent, one could still resort, as in [14], to the FM modulating data as the LMS stimulus. There, 1% of the K_{DCO} error was reported. By extending the continuity of adaptation over multiple packets, the K_{DCO} accuracy of 0.05% can be reached, which is needed to achieve a 1/3 of the maximum initial frequency error due to channel hopping via TPM ($1/3 \cdot 75\text{ kHz}/40\text{ MHz}$).

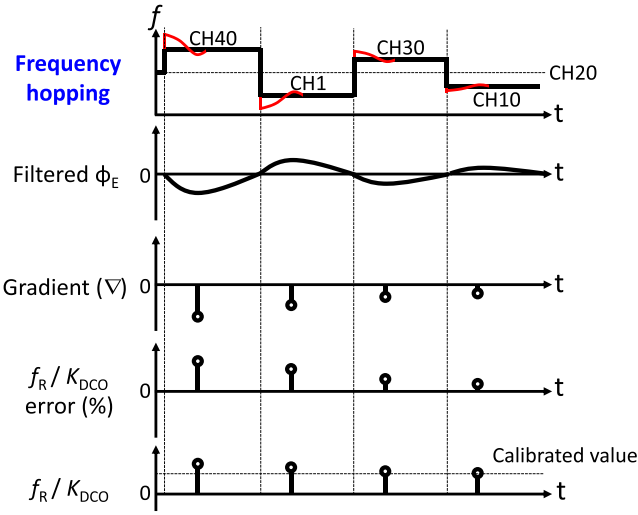
V. DCO DESIGN

Fig. 6(a) shows the proposed schematic of a class-F transformer-based DCO. The transformer turns-ratio $N = 2$ and coupling coefficient ($k = 0.74$) are designed to provide a $\sim 2\times$ passive voltage gain [17] from the drain (transformer's primary) to gate (transformer's secondary) sides of the transconductor pair, thus boosting the loop gain by 50% and improving the startup at a lower supply of 0.23 V. The DCO has five switched-capacitor (sw-cap) tuning banks: PVT, coarse (COAR), and FINE for the PLL locking onto CH20, as well as hopping (HOP1, 2) and FM tuning banks for the TPM. The PVT bank is unit weighted and provides 10.2-MHz steps.

The largest PVT capacitor bank is duplicated equally into the transformer's primary and secondary to achieve a maximum $Q(= 12)$ enhancement. To ensure enough coverage in each tuning bank, a COAR bank has a step size of 0.48 MHz



(a)



(b)

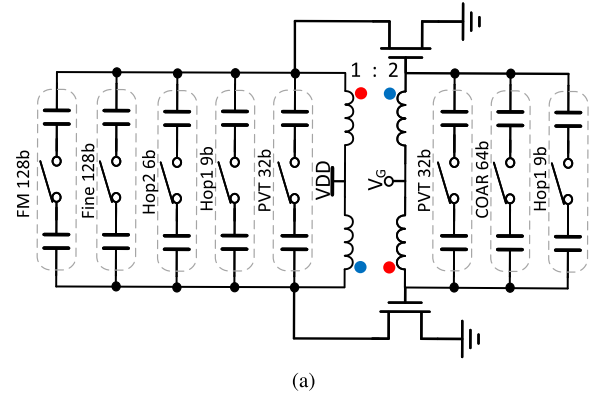
Fig. 5. (a) LMS adaptation algorithm for calibrating the DCO gain. (b) DCO gain adaptation via hopping perturbations.

and the FINE bank has the finest resolution of 16 kHz to eliminate the need for power hungry $\Sigma\Delta$ DCO dithering. To achieve the finest resolution, the FINE bank is connected to the primary coil to benefit from the capacitance transformation of $1/N^2$. The FM sw-cap bank with 128 unit-weighted 16-kHz units covers the ± 250 -kHz GFSK modulation range.

The HOP1/HOP2 banks with 14 binary-weighted bits (i.e., 8+6) are used for the fast hopping with a tuning range of 180 MHz to cover the BLE band. A binary-weighted arrangement was chosen rather than unit-weighted as it allowed reducing the occupied switched-capacitor bank area by $\sim 90\%$. However, the binary scheme produces process-related mismatch errors so it requires a compensation technique. The DCO gain for each tuning banks is shown in Fig. 6(b).

VI. NONLINEAR DCO GAIN

A previous version of this oscillator [18] has revealed a strong transfer function nonlinearity in the DCO gain, K_{DCO} . In the past, this would not be an issue since K_{DCO} is normally estimated in real time, e.g., before each TX packet.



FM	128b	unit-weighted	16kHz/bit
FINE	128b	unit-weighted	16kHz/bit
COAR	64b	unit-weighted	0.48MHz/bit
PVT	32b	unit-weighted	10.2MHz/bit
HOP1	8b	binary-weightd	89MHz(MSB)
HOP2	6b	binary-weightd	11kHz(LSB)

(b)

Fig. 6. (a) Transformer-based DCO with multi-capacitor bank for ADPLL. (b) DCO gain for each tuning banks.

However, this requires a reasonably wide ADPLL loop BW to work out any resulting disruptions. In this section, we derive a theoretical formula between the nonlinear K_{DCO} and the oscillating frequency, f , which will ensure precise channel hopping operation.

A. DCO Gain Over Frequency

The resonant frequency of an LC tank is tunable according to

$$f(L, C) = \frac{1}{2\pi\sqrt{LC}}. \quad (3)$$

Typically, the inductor L is constant (although not precisely known) so let us assume that the resonant frequency f is changed only by varying the capacitance C , i.e., $f(C)$. This rate of change is

$$\frac{df(C)}{dC} = -\frac{1}{2} \frac{f(C)}{C} = -2\pi^2 f^3(C)L. \quad (4)$$

For small capacitance changes, the resonant frequency change will be $\Delta f(C) \approx (df(C)/dC)\Delta C$. Expanding it with (4) and (5) yields

$$\Delta f(C) = -\frac{1}{2} \frac{\Delta C}{C} f(C) = -2\pi^2 (L \cdot \Delta C) f^3(C). \quad (6)$$

The above-mentioned relationship (6)–(7) ties the frequency step Δf of a DCO to the switchable capacitance ΔC . For a stable PVT, both L and ΔC are constant physical parameters as they are invariant (to the first order of approximation) with the oscillating frequency f . The right-hand side of (6)

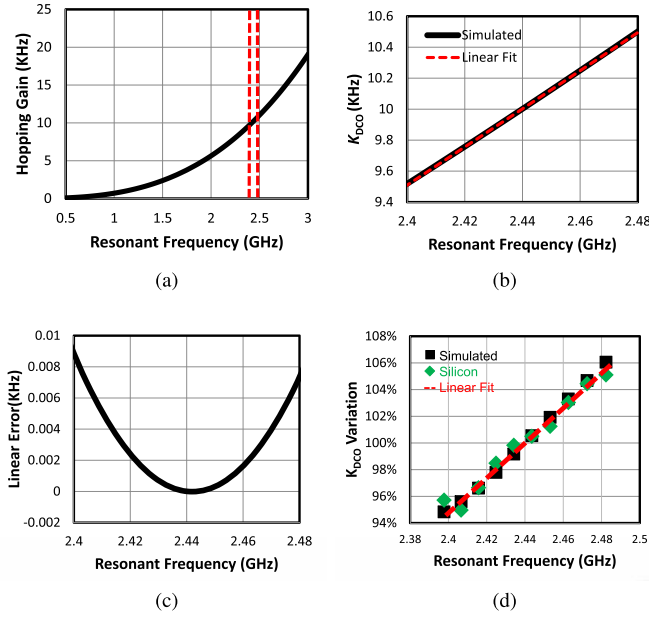


Fig. 7. K_{DCO} versus frequency. (a) Wide range, (b) over the BLE band, (c) K_{DCO} error over the BLE band, and (d) good agreement between circuit-level simulations, linear fit, and silicon results.

contains one independent variable (i.e., C) and one dependent variable (i.e., f). Hence, it is not as convenient as (7) which involves only one dependent variable (i.e., $f(C)$). Moreover, in a settled PLL, the oscillating frequency $f(L, C)$ is known precisely (at least to within the accuracy of its frequency reference, FREF, which should be very stable). This implies the precise knowledge of the LC product but not necessarily its constituent parts. For this reason, since L and ΔC are invariant with f , they are grouped together in the parenthesis in (7).

From the pragmatic point-of-view of frequency synthesis, it is more convenient to think of f as an independent variable—it is known and set precisely by tuning the imprecise C (which resonates with the imprecise L) via a closed-loop PLL system stabilized by FREF. Since $K_{\text{DCO}} \equiv |\Delta f|$ per unit switchable capacitor, (7) is rewritten as

$$K_{\text{DCO}}(f) = 2\pi^2(L \cdot \Delta C)f^3. \quad (8)$$

Taking a derivative of K_{DCO} with respect to the oscillating frequency results in

$$\frac{dK_{\text{DCO}}(f)}{df} = 6\pi^2(L \cdot \Delta C)f^2 = 3\frac{K_{\text{DCO}}(f)}{f}. \quad (9)$$

The K_{DCO} nonlinearity is conveniently termed “compression” due to a physical observation that as ΔC unit capacitors are being added to the tank (i.e., $\sum_1^k \Delta C$), the total C increases whereby the resonating frequency f decreases. Each new Δf will be smaller than the previous one.

The cubic relationship in (8) is graphically shown in Fig. 7(a) over a very wide oscillating frequency span. For implementational convenience, the LC -tank operates at double the desired frequency. This plot is fully described by $L = 2.13$ nH and $\Delta C = 4.2$ aF resulting in $K_{\text{DCO}} = 10$ kHz at $f = 2440$ MHz.

The cubic relationship is graphically shown in Fig. 7(b) over the frequency span of the Bluetooth band. Also shown in Fig. 7(c) is a linear interpolation in order to demonstrate that the curvature extent over the entire frequency range has a maximum error of merely 9 Hz.

B. DCO Gain Compensation

Equation (8) can be rewritten as for $f = f_{\text{ch}}$

$$K_{\text{DCO}}(f_{\text{ch}}) = K_{\text{DCO}}(f_{\text{ch20}}) \cdot \left(\frac{f_{\text{ch}}}{f_{\text{ch20}}}\right)^3 \quad (10)$$

$$= K_{\text{DCO}}(f_{\text{ch20}}) \cdot 1/X(\text{ch}). \quad (11)$$

It relates the DCO gain at any channel to that obtained and carefully maintained at CH_{20} via a cubic scaling factor, $X(\text{ch})$. Since the DCO gain normalizing multiplier appears of greater interest

$$f_R/K_{\text{DCO},\text{ch}} = X(\text{ch}) \cdot f_R/K_{\text{DCO},\text{ch20}} \quad (12)$$

where $K_{\text{DCO},\text{ch}} \equiv K_{\text{DCO}}(f_{\text{ch}})$ for ease of notation. However, that is of little practical consequence as the physical multiplier has to stay pinned to CH_{20} . Instead, it is the following OTW that gets adjusted by $X(\text{ch})$, as indicated in Fig. 8. Equations (1) and (2) are now rewritten to account for the dependence of K_{DCO} on frequency f such that the effective DCO gain will be flat over the channels

$$\text{OTW}_{\text{FM}} = X(\text{ch}) \cdot \text{FCW}_{\text{FM}} \cdot f_R/K_{\text{DCO},\text{ch20}} \quad (13)$$

$$\text{OTW}_{\text{ch-20}} = X(\text{ch}) \cdot \text{FCW}_{\text{ch-20}} \cdot f_R/K_{\text{DCO},\text{ch20}}. \quad (14)$$

Since the BLE band range is narrow, $X(\text{ch}) = (f_{\text{ch20}}/f_{\text{ch}})^3$ should be replaced by a linear fit, as Fig. 7(b) suggests. The derivative of $X(\text{ch})$ with respect to f_{ch} is $(dX(\text{ch})/df_{\text{ch}}) = -3 \cdot X(\text{ch})(1/f_{\text{ch}})$. Its value at f_{ch20} is $-3/f_{\text{ch20}}$. Hence,

$$X(\text{ch}) \cong 1 - 3\frac{f_{\text{ch}} - f_{\text{ch20}}}{f_{\text{ch20}}} \quad (15)$$

Fig. 7(d) shows a good agreement between circuit-level simulations, linear fit, and silicon results. If the frequency increases by 1.64% from CH_{20} to CH_{40} , the DCO gain increases by 4.9%.

C. DCO Gain Compensation for Binary Mismatch

There are actually two compensation schemes for the K_{DCO} nonlinearity, as shown in Fig. 8. The FCW is pre-compensated by: 1) aforementioned cubic compensation via $X(\text{ch})$ for the “global” nonlinearity of low spatial frequency and 2) binary “local” compensation for a mismatch of high spatial frequency [30] of the hopping bank sw-caps (see Fig. 6) via a lookup table. The blue line shows the K_{DCO} curve without the binary mismatch compensation of the hopping. The red line shows the curve with no cubic compensation but with the binary error compensation. The black line shows a linear fit for K_{DCO} . The $X(\text{ch})$ in (15) shows a reverse curve related to K_{DCO} . After multiplying the OTW_{PRE} by $X(\text{ch})$, OTW_{hop} will see an effectively flat K_{DCO} over the BLE frequency range except for the binary mismatch.

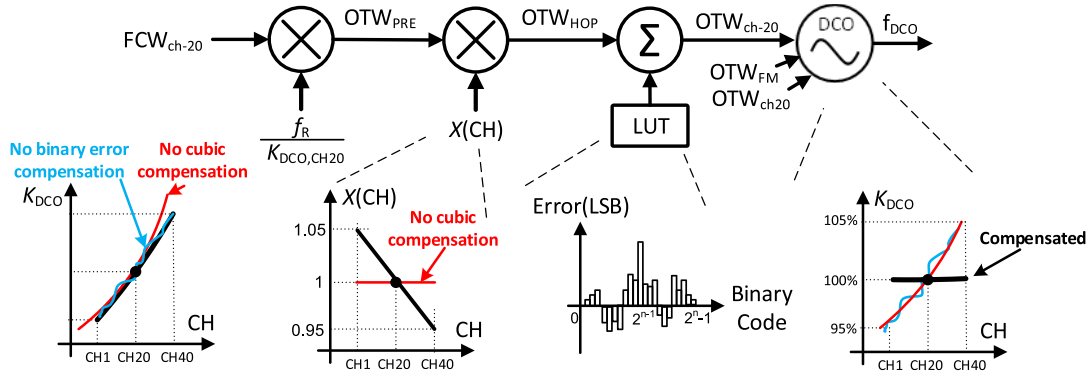
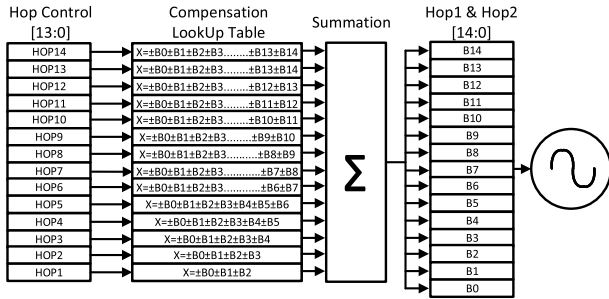
Fig. 8. K_{DCO} nonlinearity compensation schemes.

Fig. 9. Binary error compensation flow for hopping bank.

In this design, the hopping bank needs to cover the 80-MHz tuning range and reach a minimum 11-kHz resolution. If unit-weighted (thermometer) control was adopted, it would require 8000 uniform switched capacitors and excessive number of routing lines for the connections. It would cause extra area and power consumption for the extra parasitic loading. A binary control mechanism is therefore preferable for this application. However, the BLE requirement for a frequency drift is as tough as 75 kHz and so the ultralow BW operation ($BW = 2^{15}/2\pi \cdot 1/4 = 1.3$ kHz, with $\alpha = 1/2^2$ [19]) could not possibly help with locking to the right frequency during the short 250- μ s packet. Therefore, the binary error compensation for the systematic mismatch is introduced. Fig. 9 shows a detailed implementation from the 8+6 bits of the control signal to the DCO. Each controlled hopping bit will have a compensating correction (derived from silicon measurements) to be stored in a lookup table. After the summation of all the controlled bits, an optimized FCW could replace the original hopping control to the DCO. The sub-LSB, i.e., LSB-1, is designed for 5.5 kHz such that it is $\sim 10\times$ smaller than the frequency drift specification. Because the sw-caps are composed of metal-oxide-metal (MOM) capacitors of same metal layers, but of necessarily different sizes, the binary mismatch error will be mostly systematic with low random mismatch components. The systematic component is characterized in the lab using a technique adopted from [31] and then used in a lookup table for each integrated circuit device.

The MOM sw-cap sizes of the binary hopping bank are chosen to keep the random mismatch sufficiently low as not

to violate the channel hopping frequency error. The analysis is based on the commonly accepted MOM mismatch coefficient of $0.3\%/\sqrt{C}$ with C in fF [30]. The MSB-1 sw-cap value of 68.8 fF, which produces ~ 40 -MHz step size (according to (8), while considering the 4:1 impedance transformation of 1:2 step-up transformer in Fig. 6), features mismatch of only $\sigma_C = 24.9$ aF or $\sigma_f = 15.2$ -kHz standard deviation. It could be proven that the mismatch is the same whether the MOM cap of 40 MHz-weight switches on (“1000...”) or whether it stays off and all MOM caps of lower weight turn on (“0111...”).

VII. ADPLL ARCHITECTURE

A complete block diagram of the proposed ADPLL is shown in Fig. 10. It highlights supply voltage partitioning. The DCO is supplied by the lowest voltage of 0.23 V. The high-speed buffer and $\div 2$ divider are supplied by the highest voltage of 0.45 V. The digital logic of all “black” blocks can operate at 0.4 V when clocked at $f_R = 32$ kHz. The new paradigm, marked in bold, highlights the RTC as well as the TPM for frequency hopping and data modulation by applying FCW_{ch-20} and the down-sampled running average of FCW (FCW_{FM_AVG}) on the compensation path, and the corresponding OTW on the forward path. Because the 32-kHz reference clock is much slower than the modulating data, DCO is modulated directly by the sampling clock f_{FM} . On the compensation path, FCW is accumulated at rate f_R and the FCW_{FM_AVG} changes within a period of f_R to match the DCO change. Closed-loop locking on CH_{20} is achieved by applying FCW_{ch20} to the main ADPLL [1].

Cutoff frequency, f_T , of this technology is in hundreds of gigahertz, so the devices which run at the 32-kHz clock can operate at sub-threshold to directly interface with, e.g., solar cell harvesters going below 0.5 V. The only exception is the 128-stage time-to-digital converter (TDC). The unit delay in the TDC, which determines its resolution, is inversely proportional to supply voltage, so the TDC supply ($V_{2X} = \sim 0.7$ V) is boosted to twice the input voltage ($V_X = \sim 0.35$ V) by a switched-capacitor main doubler [8]. This way, the TDC supply will be close to the 0.75-V nominal level of digital logic. Consequently, a tiny voltage doubler shown in Fig. 11 is added. Switches (S1~S4) are toggled by drivers to realize a two-phase switched-cap operation.

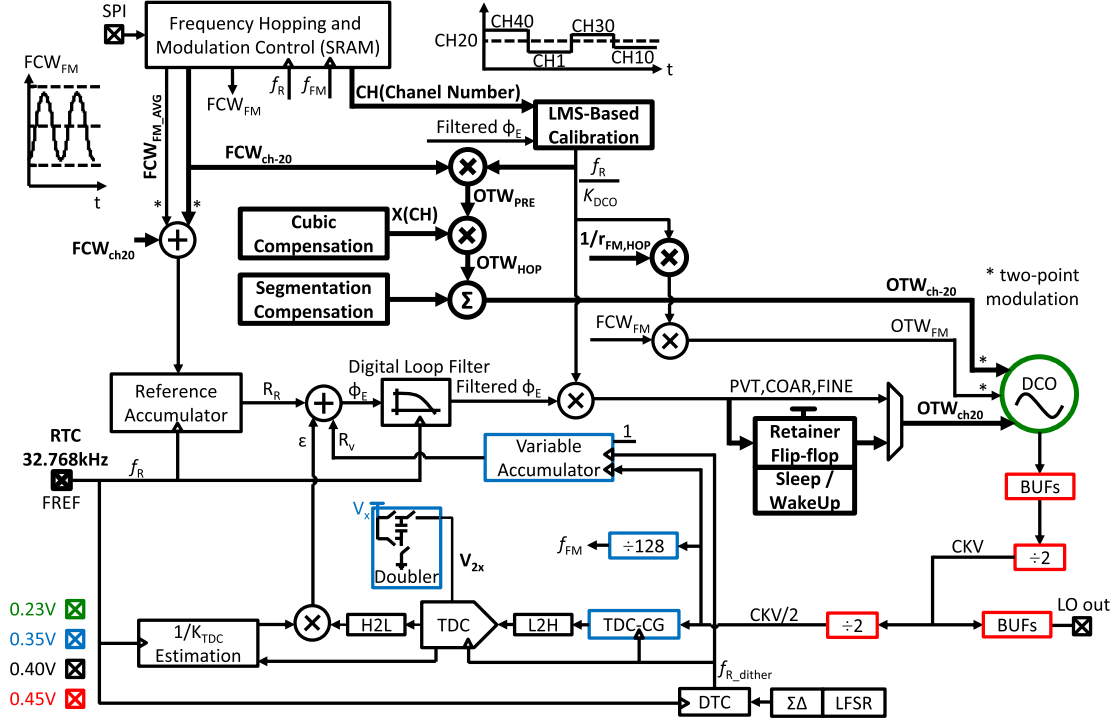


Fig. 10. Block diagram of the proposed ADPLL.

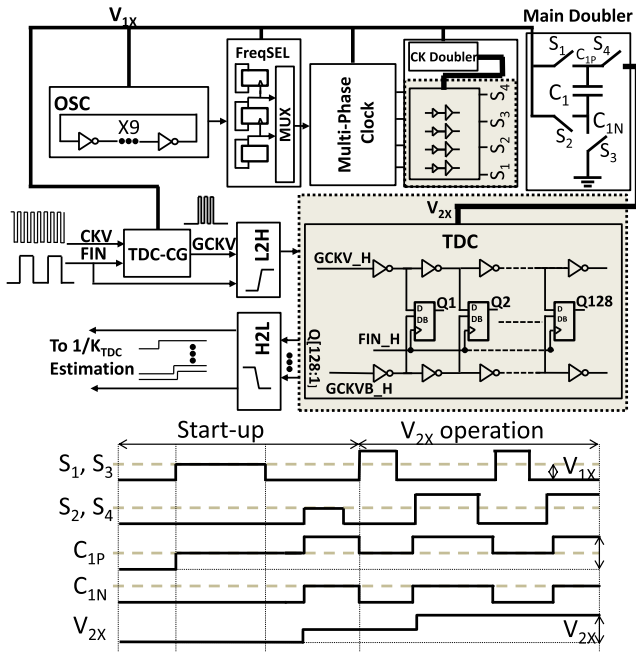


Fig. 11. Block diagram of TDC and doubler.

Voltages at C_{1P} and C_{1N} increase progressively and the output voltage at V_{2X} is boosted $2\times$ after few oscillation cycles in the startup period. The switching clock is generated by a current-starved oscillator (OSC), frequency dividers with multiplexers (FreqSEL), and a multi-phase clock generator, which are supplied by V_X . Moreover, the necessary level shifters

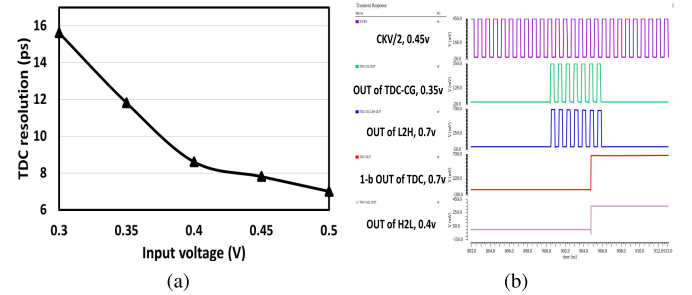


Fig. 12. (a) Measured TDC resolution versus doubler's input voltage. (b) Simulated signal waveforms in different voltage domains.

(L2H and H2L) for the $1\times \Rightarrow 2\times$ voltage domain transfers propagate signals (GCKV_H, FIN_H, and Q1~Q128) between different voltage domains. The TDC resolution is enhanced to 11.8 ps at 0.35 V of doubler input, as shown in Fig. 12(a).

The DCO clock gating scheme is adopted to reduce the power consumption of inverter chain [4]. A clock-gating (TDC-CG) circuitry cuts unnecessary edges of CKV/2 before feeding them to TDC. The DCO variable clock (CKV) of ~ 2440 MHz is divided by 256 to generate 9.53 MHz for the frequency modulation clock f_{FM} . Other clocks can be readily generated by an integer or fractional division of CKV/2.

A low duty-cycle operation requires periodical sleeps during which the ADPLL is shut down. Since the ADPLL only locks to CH₂₀, the internal closed-loop states of OTW, K_{DCO} normalization value, and integral part of loop filter are saved

in retainer registers in order to quickly restore the same center-channel frequency at wakeups [19], [28], [29].

VIII. ULTRALOW VOLTAGE DESIGN

Since the DCO consumes most of the ADPLL's power, we leverage our previous ULV design [21] to reach the sub-milliwatt power consumption of ADPLL. A trifilar design could help to provide a large passive voltage gain and allow the supply voltage well below the threshold voltage (V_t) to <0.2 V. However, the three coils require three identical capacitors for a maximum Q -factor enhancement. More power is burned due to the extra hopping bank for a large switch capacitance. In this design, the third coil is removed to further save power. Although it makes the supply voltage a bit higher than 0.2 V, it still reaches the lowest supply voltage among all the blocks. A tiny dc-dc voltage booster, which is not capable of delivering any power, was built in [21] to provide a sufficiently high control voltage for the switched capacitors. In this design, however, the FM bank toggles at ~ 9 MHz, rather than at 32 kHz for the other banks, and consumes tens of microampere for the switched capacitors. If the voltage doubler needs to supply both the TDC and the switch control voltage in DCO, the storage capacitor would have to be enlarged to provide sufficient charge and prevent large supply ripple which could induce frequency modulation of the DCO. A low- V_t device in this technology is switchable under a 0.4 -V digital supply and is adopted in the switched-capacitor to eliminate the requirement for the voltage doubler. The buffers right after the DCO, $\div 2$ divider, high-speed digital blocks of variable accumulator and TDC-CG are designed in ultralow- V_t devices to prevent a degradation of transition edges under 0.45 V. The simulation waveforms in the clock signal flow from CKV/2 to the output of H2L are shown in Fig. 12(b).

The 32 -kHz digital blocks are laid-out by auto-place and route (APR). The hold time constraint is a more critical timing concern than the setup time. The sub-threshold operation increases the rise/fall times of logic gates exponentially. On-chip variation (OCV) is increased to cover the timing variation due to this ULV operation. From the ULV design point-of-view, low- V_t devices are chosen in the 32 -kHz digital circuitry. However, if strictly considered the tradeoff between the low supply voltage and leakage current for a minimum power consumption, standard- V_t devices could be preferred.

IX. EXPERIMENTAL RESULTS

Fig. 13 shows the die photograph of the proposed ADPLL with two-point hopping/modulation capability. It is implemented in Taiwan Semiconductor Manufacturing Company (TSMC) 16 -nm CMOS FinFET technology. The total core area, including the DCO, TDC, voltage doubler, and digital logic, is 0.024 mm². It is successfully verified to meet the BLE specifications through on-wafer RF probing measurements of tens of dies. The measurement data shown in this section are representative.

Fig. 14(a) shows the nearly identical GFSK spectra on three extreme channels (i.e., first, middle, and last) and down-converted to baseband. The GFSK mask specifications are

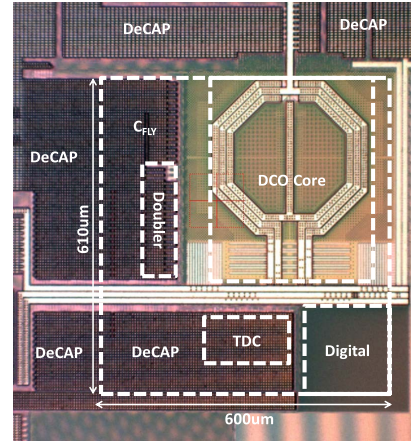


Fig. 13. Chip micrograph.

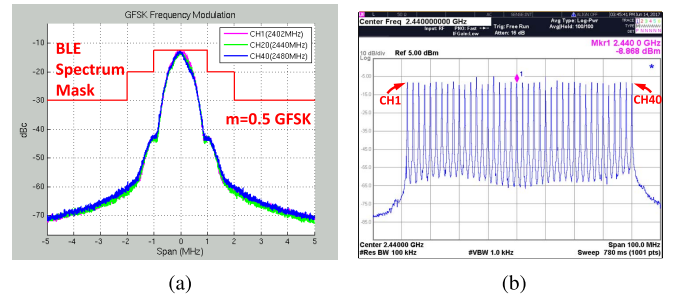


Fig. 14. (a) Measured GFSK modulated spectrum. (b) Full-band hopping across 40 channels.

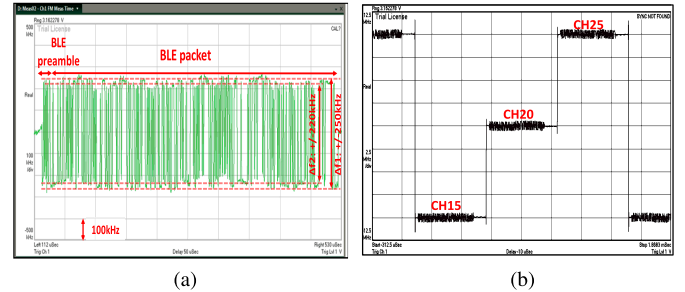


Fig. 15. (a) Measured demodulated TX BLE packet and its frequency deviation and (b) hopping among three channels.

easily met. Fig. 14(b) shows the full-band spectrum of all 40 channels at maximum hold in a random hopping sequence. Clean spectra without any smearing indicate that the instantaneous hopping is agile.

Fig. 15(a) shows the measured demodulated TX BLE packet. Stable frequency deviations between symbols “0” and “1,” ideally separated from the carrier by ± 250 kHz with no ISI [16]: Δf_1 (without ISI) and Δf_2 (with maximum ISI) demonstrate that the frequency modulation is precise and the ADPLL loop is locking well even with the 1.3 -kHz PLL BW.

Fig. 15(b) shows the time-domain measurement of sequential hopping between channels 15, 20, and 25. The GFSK modulation is mostly turned on but the intervals of nonmodulated carrier before and after each data packet are clearly visible. The y-axis shows the frequency separation. The x-axis is time, indicating the extended packet length is ~ 0.5 ms.

TABLE I
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART BLE PLLS

	This work	[10] ISSCC'18	[6] JSSC'17	[9] ISSCC'17	[4] ISSCC'14	[3] ISSCC'13	[2] ISSCC'12
Architecture	ADPLL TDC	ADPLL DTC	ADPLL TDC	ADPLL TDC+DTC	ADPLL TDC+DTC	ADPLL TDC	Analog CP-PLL
Technology	16nm FinFET	65nm	28nm	40nm	40nm	40nm	90nm
VDD(V)	< 0.45	1	1	1	1	1.3	1.2
Reference(MHz)	0.032	26	5-40	N/A	32	26	24
Output(GHz)	2.1-2.5	2.0-2.8	2.05-2.55	1.8-2.5	2.1-2.7	2.4	1.7-2.48
RMS Jitter (ps)	1.39**	0.53	1.23	1.98	1.7	0.98	2.66
Power (mW)	0.923	0.98	1.4	0.67	0.86	4.55	1.1
FOM*	-237.5	-246	-236.7	-236	-236	-233.6	-231
Core Area (mm ²)	0.24	0.23	0.24	0.18	0.2	0.075	0.75
Channel Hopping Settling Time(us)	< 0.1	N/A	15	11	N/A	N/A	< 40
TDC Resolution(ps)	7.8@0.45v 11.8@0.35v	2***	12	N/A	N/A	7	N/A

* FoM=10*log[(σ^2_{jitter})*(P_{DC}/1mW)] ** Integrated from 100kHz to 1GHz *** With calibration

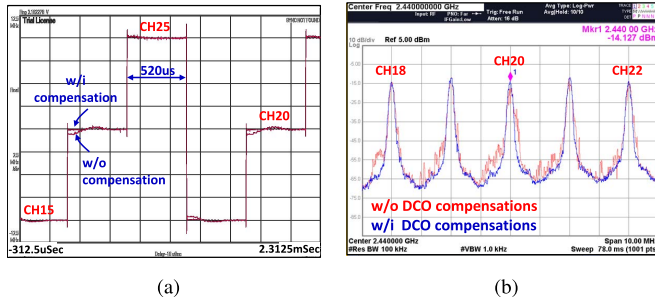


Fig. 16. (a) Settling with and without DCO compensation. (b) Measured frequency hopping in spectrum without and with DCO compensation.

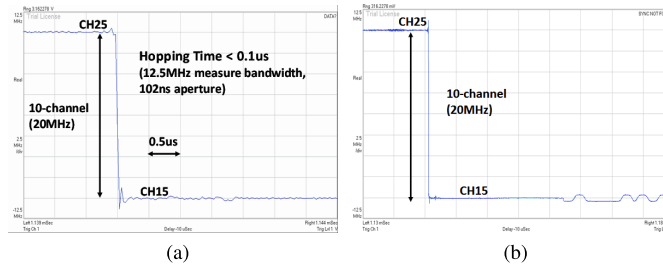


Fig. 17. (a) Measured near-instantaneous hopping and (b) its zoomed-out view.

Figs. 15(b) and 16 demonstrate the advantage of nonlinear DCO gain compensations for instantaneous hopping in both time and frequency domains. Without any compensation, larger frequency errors happen when hopping back and forth, with the resulting long settling spreading over the whole packet length. With the compensation engaged, the hopping appears instantaneous. Fig. 17(a) shows on the y-axis the demodulated TX carrier during a single hop across 10 BLE channels (20 MHz). Within the 0.1 μs time aperture of the available signal analyzer, the hopping appears instantaneous.

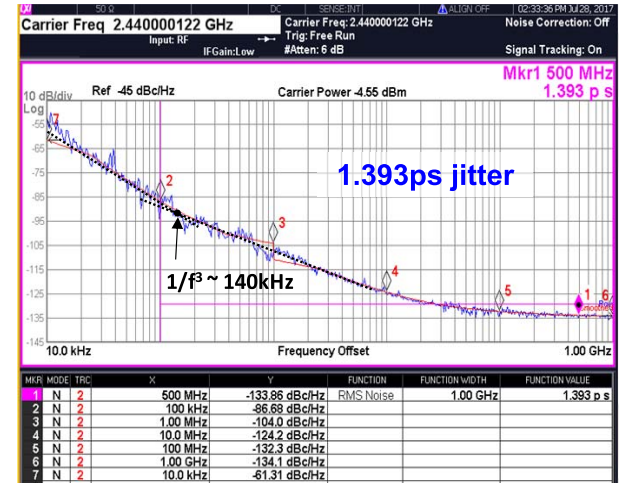


Fig. 18. Measured phase noise.

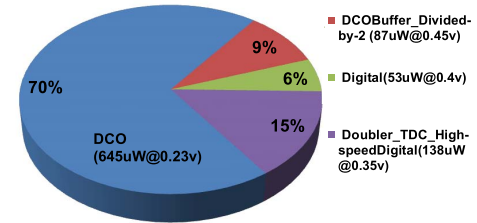


Fig. 19. Power consumption breakdown of the sub-milliwatt ADPLL.

Fig. 17(b) repeats the plot with a longer time scale which also shows the data modulation.

The measured phase noise is shown in Fig. 18. The rms jitter from 100 kHz to 1 GHz is 1.39 ps and the flicker noise corner is only 130 kHz thanks to the transformer and FinFET technology.

Fig. 19 shows the pie-chart of power consumption breakdown. ADPLL logic at 32-kHz drains the least power at

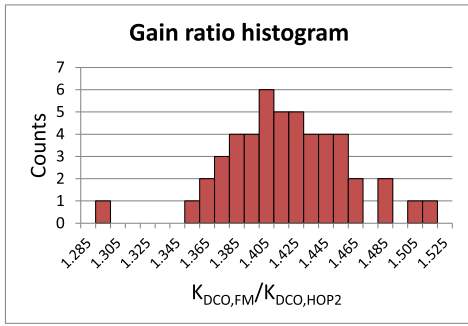


Fig. 20. Measured gain ratio histogram of $K_{DCO,FM}/K_{DCO,hop2}$.

6%. DCO consumes the most at 70%; further overall power reduction must come from the DCO. The power loss due to the crystal oscillator is conspicuously missing.

Another multiplicative correction, $1/r_{FM,hop}$, characterized in Fig. 20, is applied on the FM path to compensate for the systematic $K_{DCO,FM}/K_{DCO,hop2}$ gains ratio of 1.41 ($= r_{FM,hop}$). This gain mismatch is due to different styles and physical separation between these two types of DCO banks (unary for modulation and binary for hopping). The channel hopping uses a wide DCO tuning range of 180 MHz. In order to reduce the occupied sw-cap bank area, 14-bit binary-weighted control is used. On the other hand, only ± 250 -kHz tuning range is required for data modulation, so 128 unit-weighted control is appropriate. 49 dies worth of the silicon data was used for the analysis. The gain ratio mean value is 1.41 with 0.04 of standard deviation. In ± 3 std deviations, the modulation index is still within the spec of 0.45–0.55.

Table I summarizes the performance of the proposed ADPLL and compares it with state-of-the-art BLE PLLs. Its salient feature is the elimination of conventional XO as FREF in favor of the 32-kHz RTC. The ADPLL operates at an ULV of less than 0.45 V. It achieves near-instantaneous channel hopping, while maintaining the best-in-class performance at sub-milliwatt power consumption.

X. CONCLUSION

This paper demonstrates an ADPLL for BLE that eliminates the need for a crystal oscillator (XO) other than a 32.768-kHz RTC for IoT solutions. By eliminating the bulky XO, the system power consumption, occupied volume, and cost could be greatly saved. We propose to replace the conventional channel settling with a band settling that would be carried out only once per global device power up. ADPLL locks to the center of the BLE band upon system power-up and jointly performs channel hopping and GFSK modulation in a two-point manner to overcome a narrow PLL BW due to 32.768-kHz reference clock to achieve instantaneous channel hopping. Through the detailed derivation of DCO nonlinear gain behavior, extensive calibrations linearize the effective DCO transfer function free from the cubic curvature and binary tuning error to achieve a precise frequency range of hopping and modulation. Realized in TSMC 16-nm FinFET, it consumes sub-milliwatt at 0.45 V, while achieving state-of-the-art performance and less than 100 ns hopping time in a

random channel hopping sequence. This paper shows a high potential for the solution of low-cost, low power consumption, and fast hopping time in portable wireless applications.

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